

E-Commerce Sales Analysis Report

Data Cleaning, Exploratory Analysis, Key Findings, and Conclusion

Prepared for: Course 2 Group Project

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Date: _____

1. Project Constraints and Approach

This project was completed under a short timeline, so the data cleaning process was intentionally kept practical and focused. The goal was not to perform a full accounting-level audit of every transaction, but to clean the dataset enough to support the main business questions in the final analysis notebook.

Because of the time constraint, the team focused on the most important issues that could directly affect the analysis: duplicates, missing values, zero-value transactions, returns or cancellations, customer ID handling, and date/time formatting.

- Remove duplicate rows so sales and quantity totals would not be overstated.
- Remove zero UnitPrice rows because they did not represent normal sales transactions.
- Keep cancellation and return records because they affect net sales and the sales versus returns analysis.
- Create useful analysis columns such as Total_Sales, YearMonth, InvoiceDayName, InvoiceHour, Invoice_Time_of_Day, and CustomerID_Label.
- Use the cleaned dataset to answer focused business questions about countries, trends, customers, products, order timing, and returns/cancellations.

This approach allowed the analysis to move forward quickly while still addressing the main data quality issues that would affect the project findings.

2. Short Data Cleaning Summary

The original e-commerce dataset contained invoice-level transaction records, including invoice number, product description, quantity, invoice date, unit price, customer ID, and country. The cleaning notebook reviewed the structure of the data, checked for duplicates and missing values, and investigated records that appeared to be non-sales activity.

- Duplicate rows were identified and removed.
- Rows with missing product descriptions and zero UnitPrice values were reviewed as likely stock adjustments or non-sales records.
- Rows with UnitPrice equal to zero were removed from the final sales dataset.
- Negative Quantity rows were reviewed because they may represent cancellations, returns, or adjustments.
- Invoice numbers starting with C were treated as cancelled orders or returns.
- CustomerID missing values were not removed because those rows still contained useful sales, product, date, and country information.
- A CustomerID_Label column was created so missing customer IDs could be labelled as Unknown Customer.
- The final cleaned dataset was saved as Online_Retail_clean2.csv and used in the analysis notebook.

The cleaning was kept short because of the project timeline, but it was enough to make the dataset usable for the selected business questions.

3. Analysis Objective

After cleaning the dataset, the analysis focused on answering business questions related to sales performance, customer behaviour, product demand, order timing, and the effect of returns or cancellations. The main questions answered in the analysis were:

1. Which countries generated the highest total sales?
2. What was the overall monthly sales trend?

3. How did new customers compare with returning customers by month?
4. When were purchases made by day of week, hour of day, and time of day?
5. Which products were the best-selling overall?
6. Which products were the best-selling by country?
7. What were the top 10 largest orders, and when were they made?
8. Which known customers generated the highest total sales?
9. What was the impact of returns and cancellations compared with sales?

4. Exploratory Data Analysis

4.1 Distribution of Total Sales

The Total_Sales column was created by multiplying Quantity by UnitPrice. This value can be positive for sales or negative for returns and cancellations. The full histogram showed that most transaction values were concentrated close to zero, while a small number of very large positive and negative values stretched the x-axis.

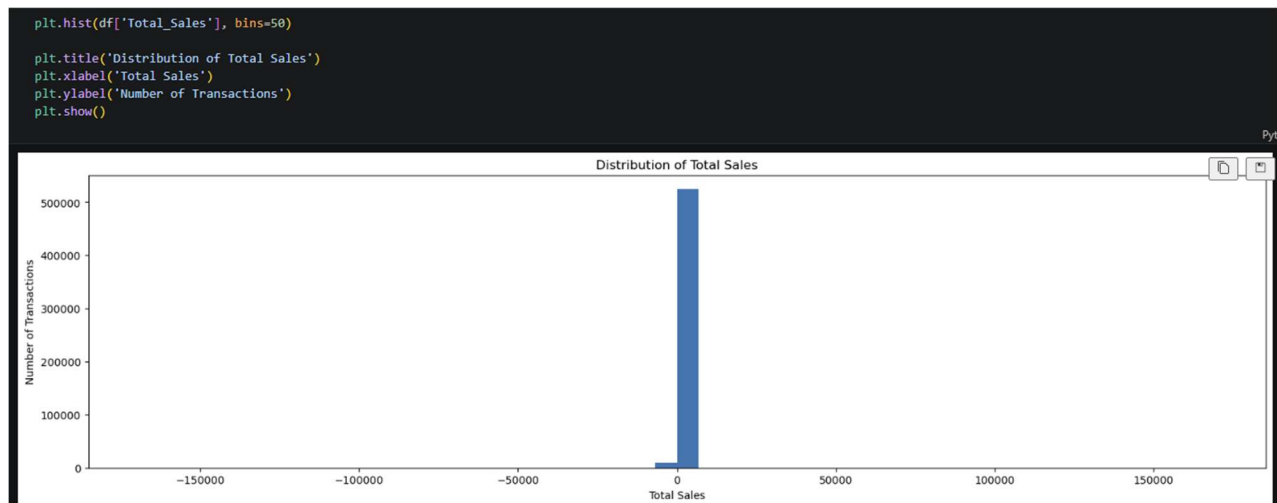


Figure 1. Full distribution of Total_Sales.

To better understand normal transaction activity, a zoomed-in histogram was created using positive Total_Sales values up to 100. This showed that most individual transactions were relatively small, with many sales below 20. The distribution is right-skewed, meaning there are many low-value transactions and fewer high-value transactions.

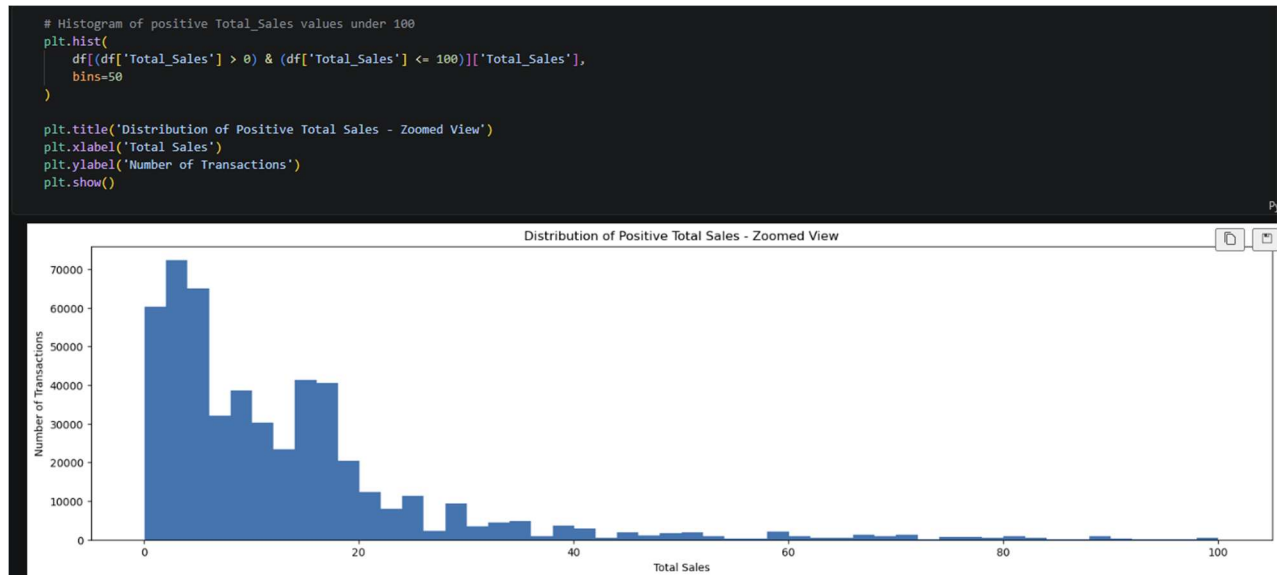


Figure 2. Zoomed-in distribution of positive *Total_Sales* values.

4.2 Box Plot of Total Sales

A zoomed-in box plot was also created using *Total_Sales* values between -100 and 500. The box plot showed that the middle range of transaction values was concentrated in a relatively small band, while many observations appeared as outliers above and below the main distribution.

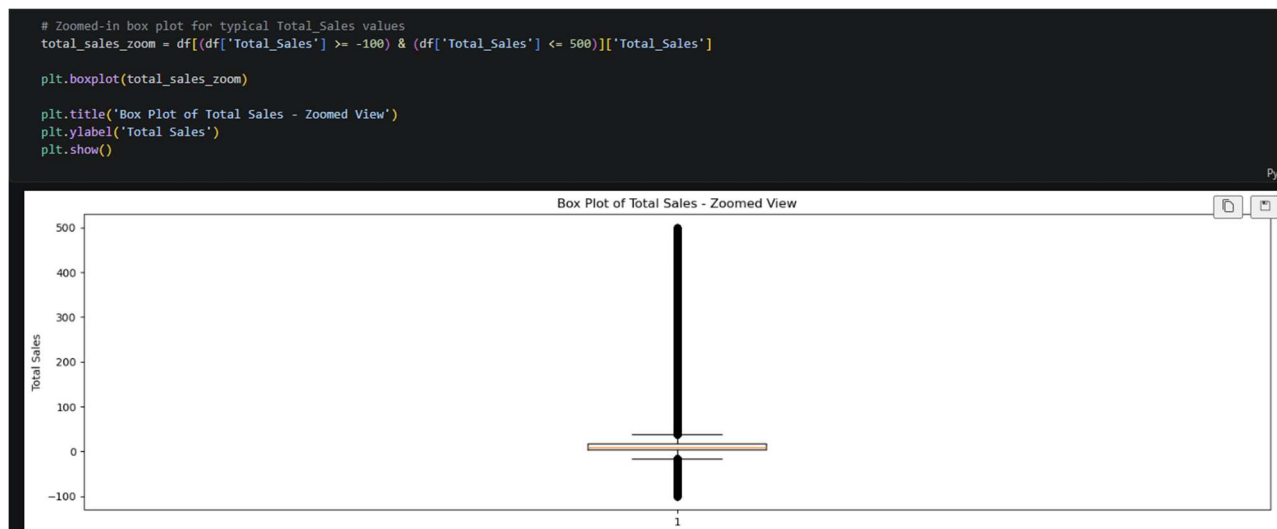


Figure 3. Zoomed-in box plot of *Total_Sales*.

This confirmed that most transaction values were modest, but the dataset also contained unusually large positive and negative transactions. These outliers helped justify reviewing large orders and returns/cancellations separately.

5. Key Business Questions and Findings

5.1 Top 10 Countries by Total Sales

The United Kingdom generated the highest total sales by a very large margin, with approximately 8.18 million in sales. The next highest countries were the Netherlands, EIRE, Germany, and France, but their totals were much smaller than the United Kingdom. This shows that the business was heavily dependent on the UK market.

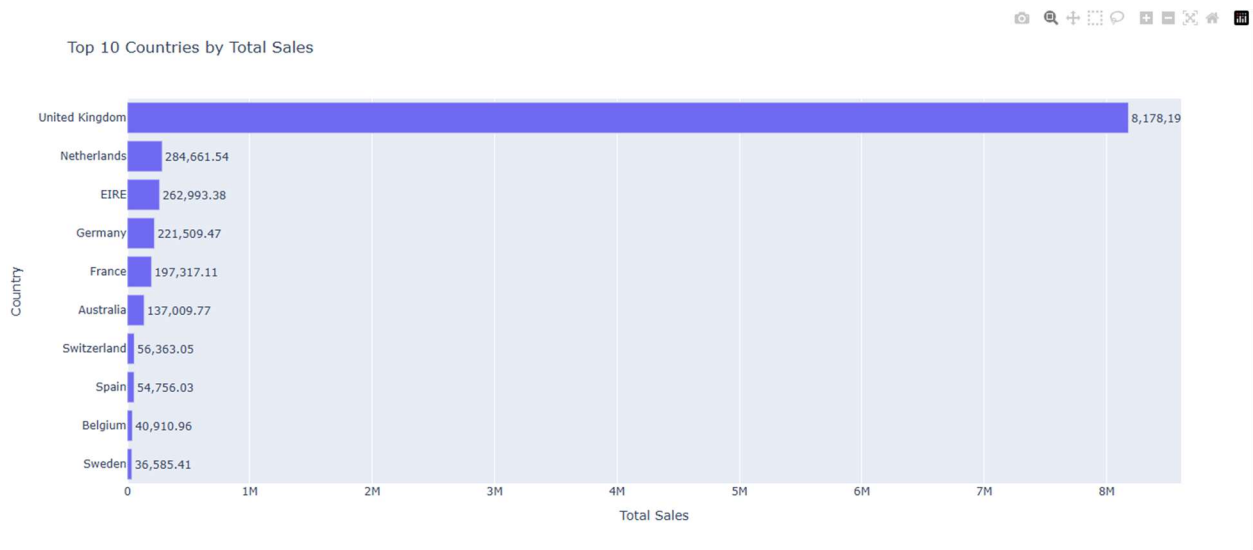


Figure 4. Top 10 countries by total sales.

5.2 Overall Monthly Sales Trend

Monthly sales fluctuated during the year, but there was a noticeable increase from September to November 2011. Sales reached their highest point in November 2011. December showed a sharp drop, which may be because the December data appears incomplete, so it should be interpreted carefully.

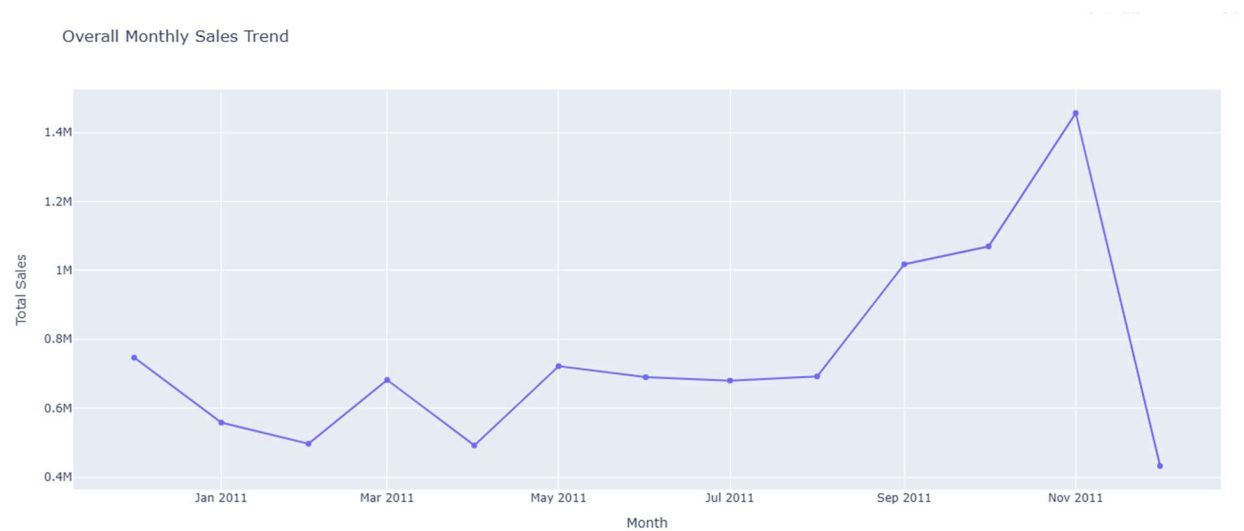


Figure 5. Overall monthly sales trend.

5.3 New vs Returning Customers by Month

Returning customers were higher than new customers in most months. This suggests that repeat customers played an important role in the business. Returning customer counts increased strongly in the later months, especially September, October, and November, which also aligns with the stronger sales trend during that period.

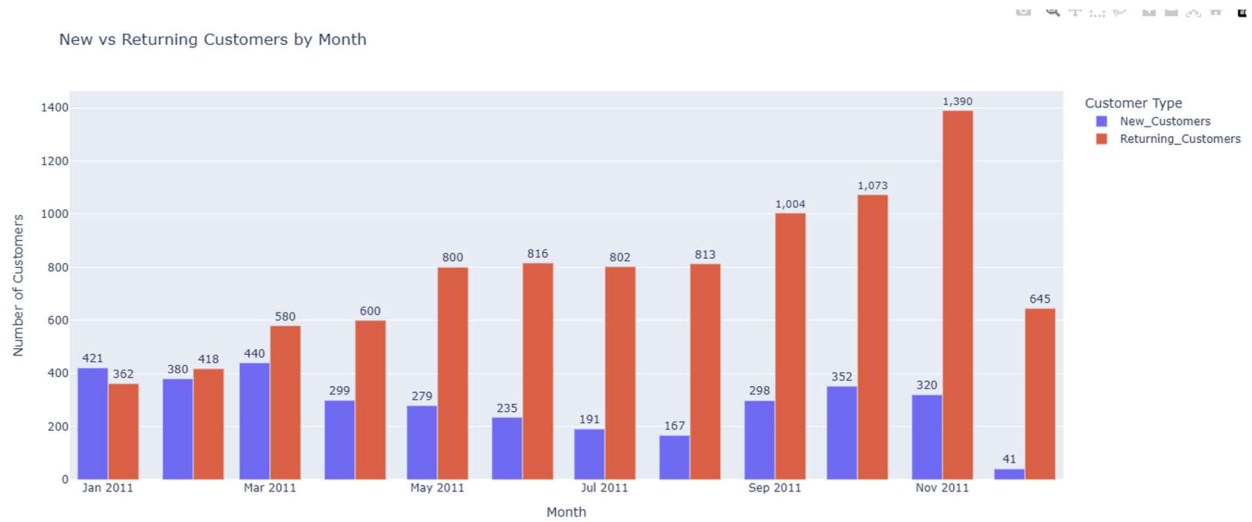


Figure 6. New vs returning customers by month.

5.4 Purchases by Day and Hour

Most purchases were made during the business week, with Thursday having the highest number of transactions. There were no Saturday transactions shown in the dataset, and Sunday had fewer purchases than most weekdays.

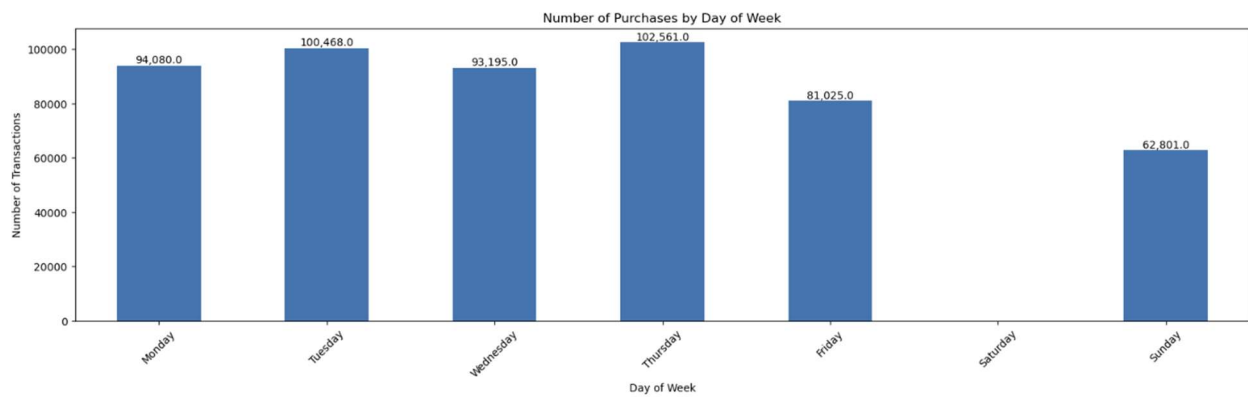


Figure 7. Number of purchases by day of week.

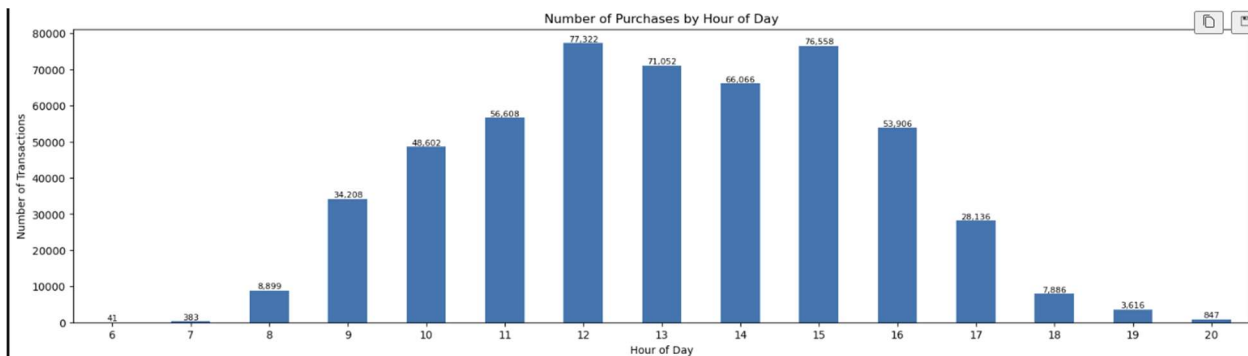


Figure 8. Number of purchases by hour of day.



Figure 9. Number of purchases by time of day.

By hour, purchases were mainly made during normal daytime hours. The highest transaction volumes occurred around 12 PM and 3 PM. When grouped by time of day, afternoon had the highest number of transactions, followed by morning and evening.

5.5 Top 10 Best-Selling Products Overall

The best-selling product overall was PAPER CRAFT, LITTLE BIRDIE, with approximately 80,995 units sold. Other high-selling products included MEDIUM CERAMIC TOP STORAGE JAR, WORLD WAR 2 GLIDERS ASSTD DESIGNS, and JUMBO BAG RED RETROSPOT. Only positive quantity records were used so that returns or cancellations did not reduce the product sales count.

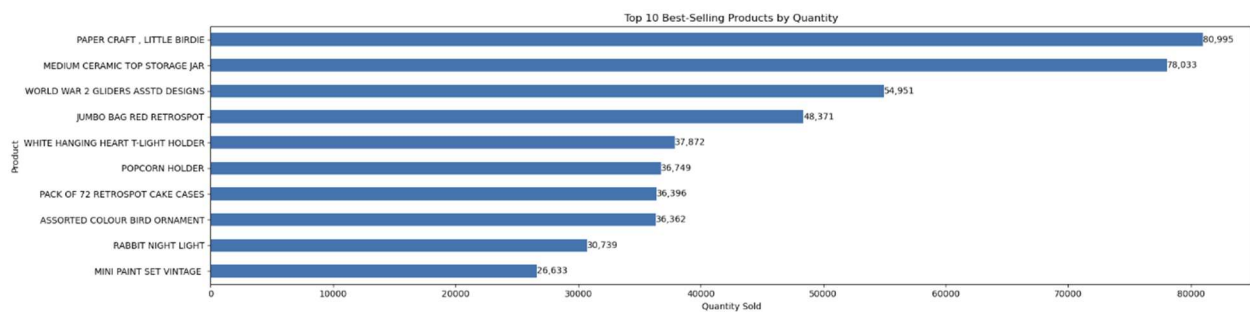


Figure 10. Top 10 best-selling products by quantity.

5.6 Best-Selling Product by Country

The best-selling product by country showed that the United Kingdom again dominated the results, with PAPER CRAFT, LITTLE BIRDIE as the top country-product combination. Other countries had much lower quantities, but the chart still helped identify popular products in different markets. For example, RABBIT NIGHT LIGHT appeared as a top product in several countries.

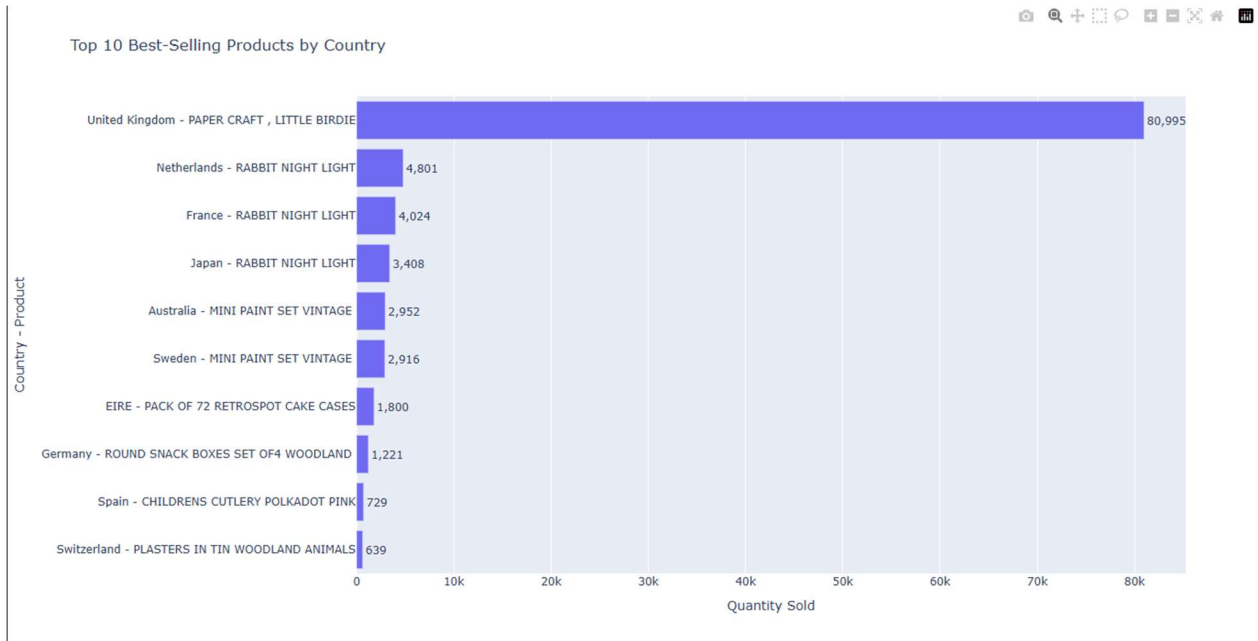
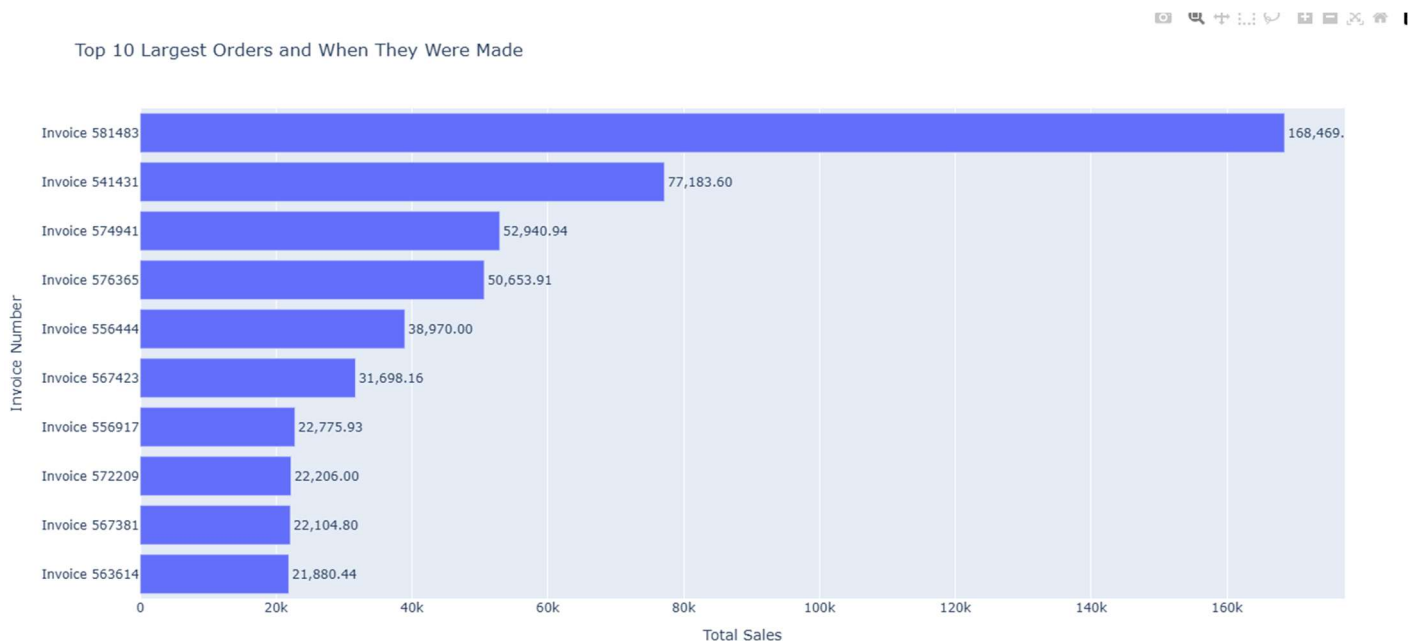


Figure 11. Top country-product combinations by quantity sold.

5.7 Top 10 Largest Orders and When They Were Made

The largest orders were reviewed by grouping transactions by InvoiceNo and summing Total_Sales for each distinct order. Additional fields such as invoice date, day of week, hour, time of day, customer label, and country were included to understand when the largest orders were made and who was associated with them.



This analysis is useful because large orders can strongly affect total sales and may need to be reviewed separately from regular transaction-level activity.

5.8 Top 10 Customers by Total Sales

Customer 14646 generated the highest total sales, followed by customers 18102, 17450, 14911, and 12415. Unknown Customer was excluded from this customer-level chart because it represents missing customer information rather than a known customer. This finding shows that a small group of known customers contributed significantly to total revenue.

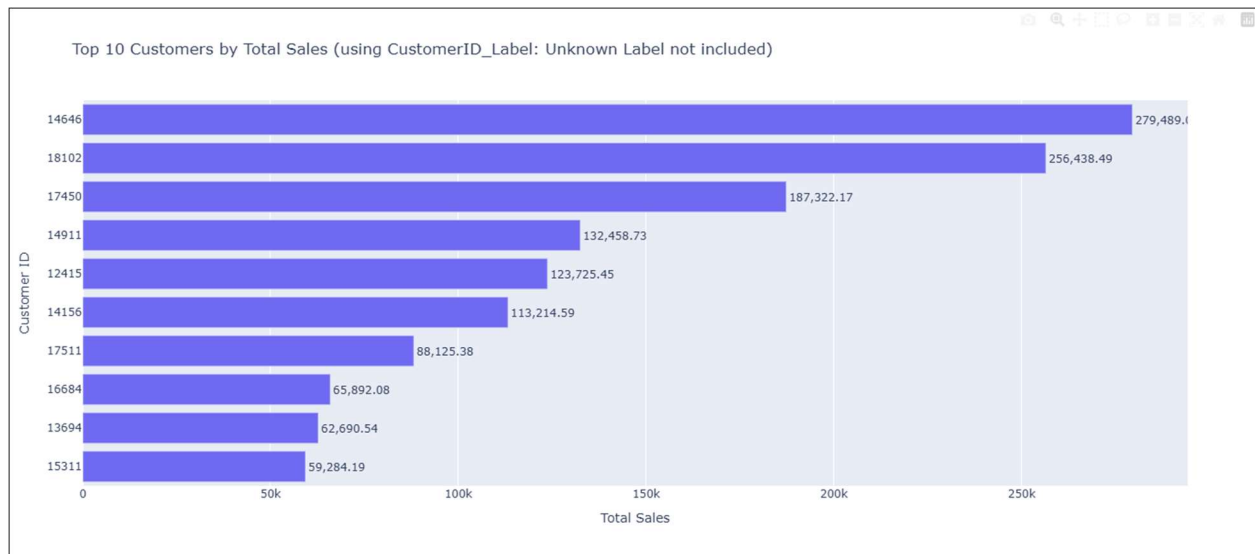


Figure 12. Top 10 known customers by total sales.

5.9 Sales vs Returns/Cancellations Impact

Positive Total_Sales values were treated as sales, while negative Total_Sales values were treated as returns or cancellations. The pie chart used absolute values so the proportional impact could be compared clearly.

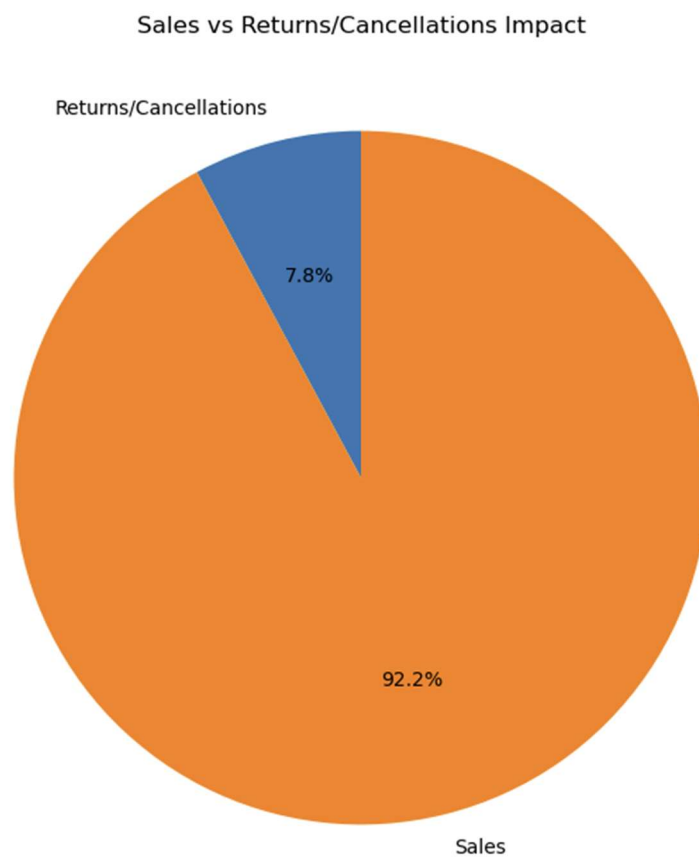


Figure 13. Sales vs returns/cancellations impact.

Sales represented approximately 92.2% of the transaction value, while returns and cancellations represented approximately 7.8%. This shows that returns and cancellations had a noticeable impact, but the majority of transaction value still came from completed sales.

6 Dashboard -Dash

To support the final analysis, an interactive dashboard was created using **Python Dash and Plotly**. The purpose of the dashboard was to present the main findings from the cleaned e-commerce dataset in a simple, visual, and interactive format.

The dashboard includes three dropdown filters:

- **Country**
- **Month**
- **Time of Day**

These filters allow users to explore the data from different angles. For example, the viewer can look at overall sales across all countries, or narrow the dashboard to a specific country, month, or purchasing period such as morning, afternoon, or evening.

At the top of the dashboard, KPI cards summarize the main business measures, including **Total Sales**, **Total Invoices**, **Known Customers**, and **Returns/Cancellations**. These cards provide a quick snapshot of business performance before reviewing the charts in more detail.

The dashboard includes four main visualizations:

1. **Top 10 Countries by Total Sales**
This chart identifies the countries generating the highest revenue.
2. **Monthly Sales Trend**
This line chart shows how total sales changed over time and helps identify stronger or weaker sales months.
3. **Top 10 Products by Quantity Sold**
This chart highlights the products with the highest sales volume.
4. **Sales vs Returns/Cancellations Impact**
This chart compares completed sales against returns and cancellations, showing how much returns reduced overall sales performance.

Overall, the dashboard helps turn the cleaned dataset and analysis results into an easy-to-understand business summary. It allows users to interact with the data and quickly answer questions related to sales performance, customer activity, product demand, and the impact of returns or cancellations.

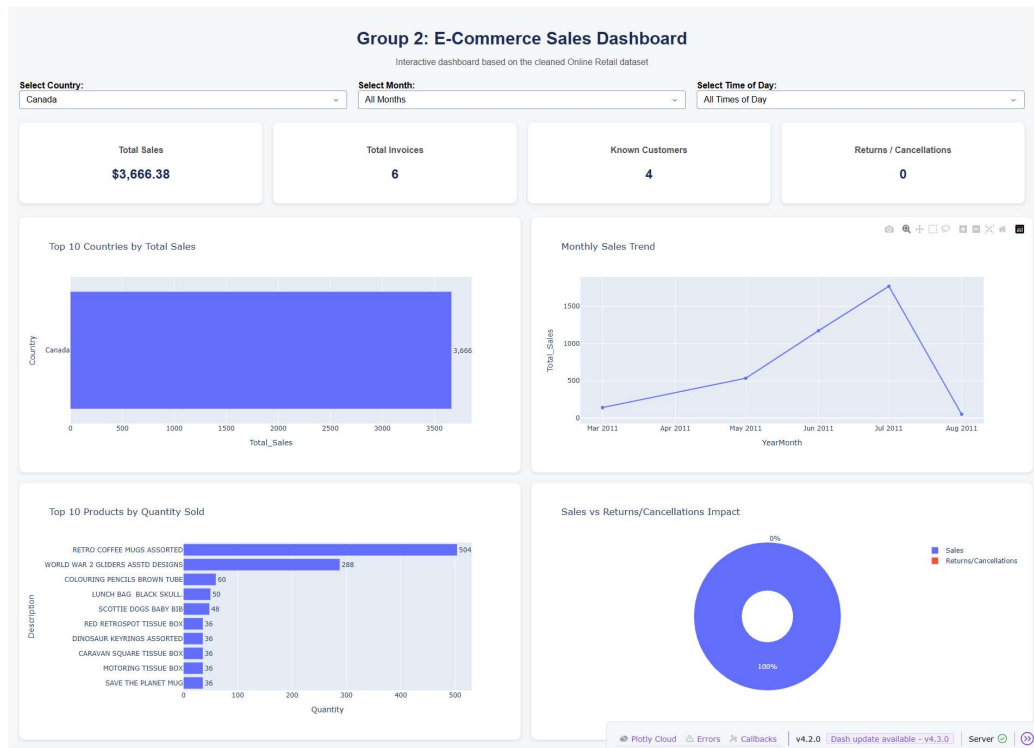


Figure 14. Screenshot of the output of the Dashboard

Conclusion

Overall, the analysis showed that the business was highly concentrated in the United Kingdom, with strong sales growth toward the end of 2011. Returning customers were an important part of the business, and purchases were mostly made during weekday daytime hours, especially in the afternoon.

The analysis also showed that a few products and customers contributed heavily to overall sales. Most transactions were relatively small, but the dataset contained several outliers, including large orders and negative sales values related to returns or cancellations.

Although the data cleaning process was kept short because of the project timeline, the main data quality issues were addressed well enough to support the analysis. Duplicate rows and zero-value non-sales records were removed, cancelled orders were identified, missing customer IDs were handled using an Unknown Customer label, and date/time columns were created for trend analysis.

The cleaned dataset allowed the team to answer key business questions related to country sales, monthly trends, customer behaviour, product performance, large orders, top customers, and the impact of returns or cancellations.

Appendix: Notebooks Used

- Final-Cleaning(1).ipynb - cleaning and preparation notebook.
- Final Analysis(2).ipynb - analysis and visualization notebook.
- Final Dashboard.py- Dashboard using Dash in Python